Plant Regrowth and Burrowing on the jenna-testing Branch

What the branch's plant regrowth, predator armour, plant-claw rule and plant-death change do to the burrowing claw and the soil, measured over 936 simulations of 100,000 ticks each.

Summary

- **With the branch's run settings** (from `run_config_tick73074.csv`), burrowing claws evolved and took over the prey in 16 of 20 runs. Every prey then died out in 4 of 20 runs. That never happened with harmless burrowing (0 of 20), and without burrowing prey survived all 20 runs.
- **Plant regrowth is the strongest effect.** With the run settings, switching it off raised the collapses from 4 to 10 of 20 runs ($p = 0.0004$ against harmless burrowing). With a second parameter set, collapses went from 4 of 20 with regrowth off, to 2 of 20 at 0.25%, to none at 1% or 3%.
- **With regrowth on, a collapse isn't the end.** In all 4 collapses under the run settings, plants regrew and new prey evolved from them; the runs ended with 48 to 145 prey. With regrowth off, 9 of the 10 collapses ended with no organisms left.
- **Dead plants leaving no food is the second effect.** With regrowth off, giving dead plants back their food raised collapses from 4 to 12 of 20 runs ($p = 0.02$).
- **Predator armour and the no-claws-on-plants rule** made no clear difference.
- **The run's graph** matches the most common outcome under its settings (9 of 20 runs): claws spread, the soil and population crash, then the claws die away and the world recovers.
- **The stay hidden range controls how far claws spread:** they took over in 6, 12 and 17 of 20 runs at ranges 3, 6 and 10.

What the branch adds

- **Plant regrowth.** A new Run Setup setting, *Plant regrowth chance*, default 1%. Each tick the world has that chance of placing a new plant at a random free spot whose soil is fertile enough for plants, which is about one plant every 100 ticks.
- **Armoured predators.** The starting predator has two armour cells: 8 cells instead of 6, so it lives longer and needs more food per offspring.
- **No claws on plants.** Claw and producer cells can't be on the same organism.
- **Dead plants leave no food.** Every organism now leaves empty cells when it dies.
- **Right-facing prey.** The second starting-prey slot places a regular prey facing right instead of a clawed prey.

How the experiments were run

Each run used the branch's own `src/` code in Node, without the browser (`analysis/headless/run_sim.js`). The headless world calls the branch's `WorldEnvironment.regrowPlant` every tick, so plant regrowth works exactly as in the browser. Every setup was run from 20 starting layouts (placement seeds 1 to 20) for 100,000 ticks; the predator-lifespan checks used 8 layouts.

Two parameter sets. The *run settings* are every value in the run config export, on its 306 by 140 grid: soil regrowth 0.01%, hides of 200 ticks, stay hidden range 10, burrowers moving every 2 ticks, free claw cells, a 0.5 chance that a clawed prey growing a killer cell becomes a predator, predator lifespan multiplier 2, plant regrowth

1%, burrowing at any predator sighting, and 10 predators, 10 + 10 prey and 39 plants. The *reference settings*, on a 288 by 180 grid, check that the conclusions don't depend on one configuration: predator lifespan multiplier 2, burrowing at any sighting, claw cells not free, every clawed killer becoming a predator, 20 prey with no claws, hides of 100 ticks, soil cost 10 per burrow, stay hidden range 6, soil regrowth 0.05% and plant regrowth 1% unless stated.

Switching changes off. Armour, the plant-claw rule and the plant-death change are built into the code, so the runner switches them off in memory for a single run (`--ablate`); the code on disk isn't changed. Plant regrowth is switched off through its setting.

Outcomes. Claws *take over* once at least 50 prey have a claw and they make up at least half of all prey. A run counts as a *collapse* if every prey dies after that, or if it ends with 2 or fewer prey and 5 or fewer plants. A *harmless burrowing* setup is the same burrowing with no soil cost; a *no burrowing* setup makes burrowing impossible, so claws can still evolve but do nothing. Differences in the number of collapses are tested with two-sided Fisher's exact tests.

Results with the run settings

Run settings, 20 runs each	Prey never died out	Claws took over	All prey died after take-over	Median tick of collapse
As exported	16 / 20	16 / 20	4 / 20	80,500
Harmless burrowing	20 / 20	15 / 20	0 / 20	–
No burrowing	20 / 20	0 / 20	0 / 20	–
Plant regrowth off	10 / 20	15 / 20	10 / 20	51,000

With the settings as exported, collapses (4 of 20) are more common than with harmless burrowing (0 of 20), but 20 runs aren't enough to be sure (p = 0.11). Switching plant regrowth off raises collapses to 10 of 20 (p = 0.0004 against harmless burrowing; p = 0.096 against regrowth on), and they come sooner (median tick 51,000 instead of 80,500). The lowest plant count in a run fell to a median of 94 with regrowth and 2 without; the soil fell to a median low of 66% of its maximum with regrowth and 58% without.

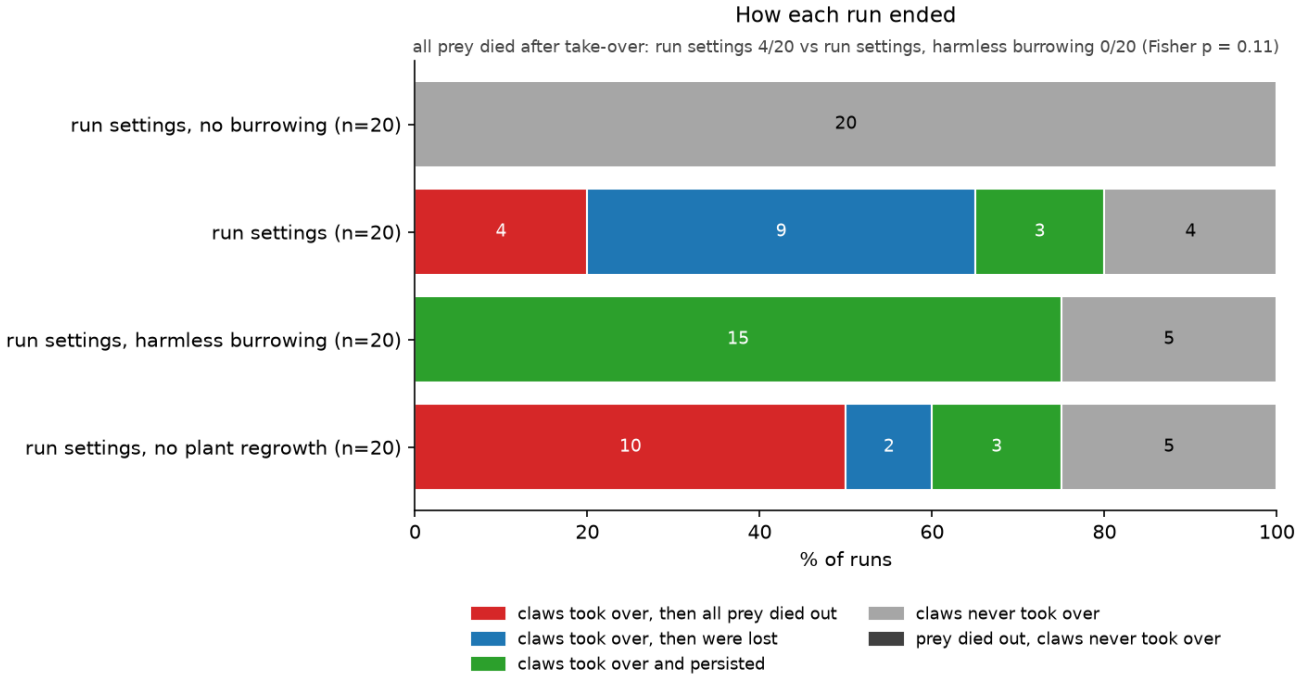


Figure 1. How each run ended with the run settings. Red: every prey died after the claws took over. Blue: the claws took over and later disappeared while prey without them carried on. Green: the claws took over and persisted. Grey: the claws never took over.

The run's graph

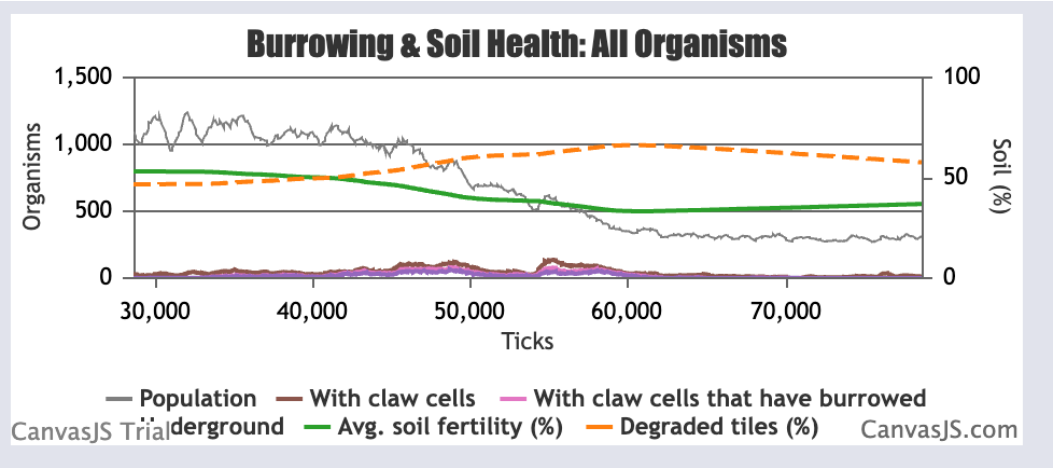


Figure 2. The browser's *Burrowing & Soil Health* chart from the original run, ticks 28,000 to 79,000. The population falls from about 1,100 to 300 by tick 60,000 while the average soil fertility drops from 53% to 33% and degraded tiles rise from 47% to 67%. Organisms with claws peak at about 130 around ticks 45,000 to 55,000, then nearly disappear, and the soil begins to recover.

Reading the chart. The chart's soil line averages fertility over the whole grid, including the permanently infertile paths, which cover 46% of this grid. Soil can therefore never read above about 54%, and 53% means the fertile soil is almost untouched. Degraded tiles are cells below half the starting average (fertility 2 or less), so the infertile paths count as degraded from the start: 47% is the baseline, and the rise to 67% means about two fifths of the fertile soil was degraded at the worst point.

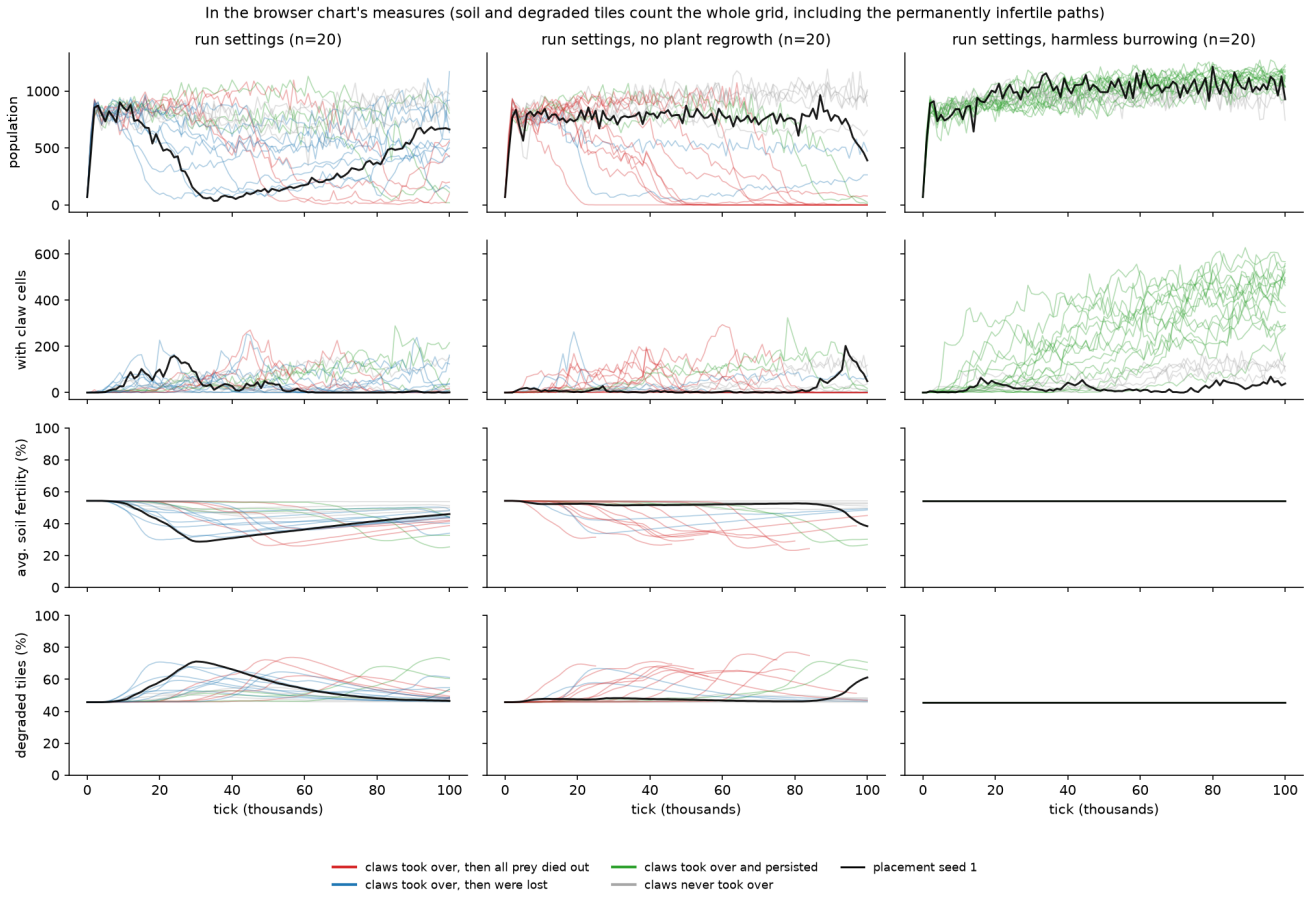


Figure 3. The 20 runs of the run settings in the chart's four measures, with regrowth on, with regrowth off and with harmless burrowing. Layout 1, the starting layout of the original run, is drawn in black. Colours as in Figure 1.

In these units the original run matches the blue runs, the most common outcome (9 of 20). The claws spread, the soil falls to 30 to 40%, the population drops by two thirds or more, then the claws die away and the soil and population slowly recover. The original run was still recovering at tick 79,000. Collapses under these settings tend to come late (median tick 80,500), so a later collapse can't be ruled out, but in these runs a claw population that had died away rarely came back. With regrowth off, the same settings more often end in a complete collapse (red), and most collapsed worlds end with nothing alive.

Plant regrowth

Regrowth places plants anywhere the soil is still fertile, so plant colonies restart away from the burrowed patches around the prey. With the reference settings, burrowing that costs the soil 10 per burrow collapsed 4 of 20 runs with regrowth off, 2 of 20 at 0.25%, and none at 1% or 3%. Harmless burrowing and no burrowing never collapsed at any rate, and without burrowing prey survived all 20 runs at every rate.

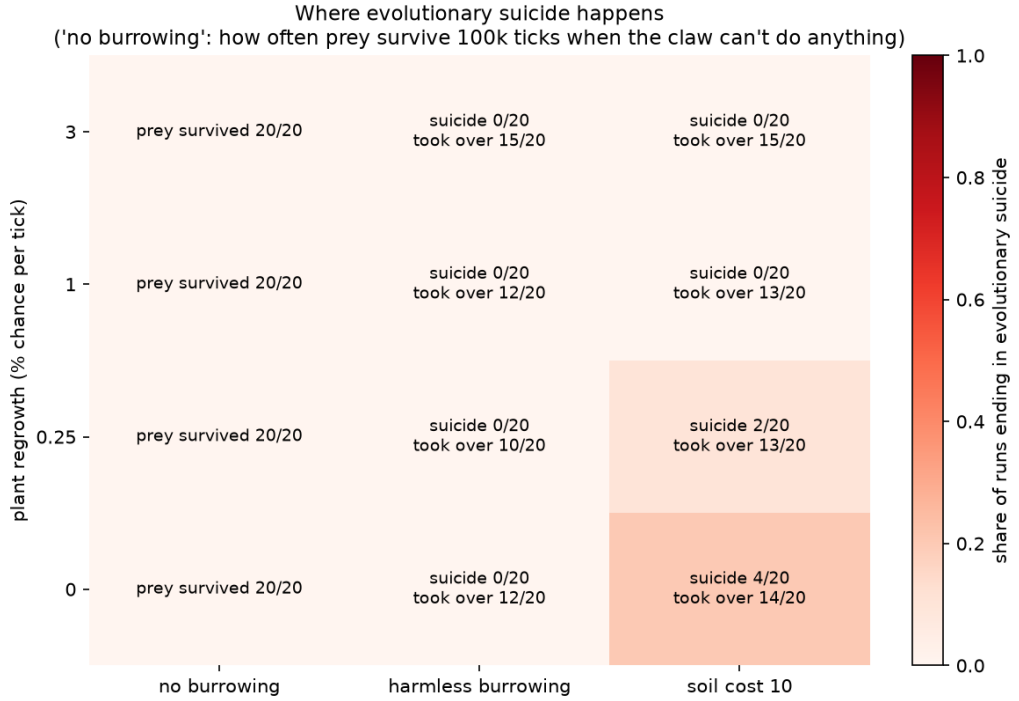


Figure 4. Share of runs ending with every prey dead after the claws took over, by plant regrowth rate, with the reference settings. Each cell is 20 runs. The *no burrowing* column shows how often prey survive when the claw can't do anything.

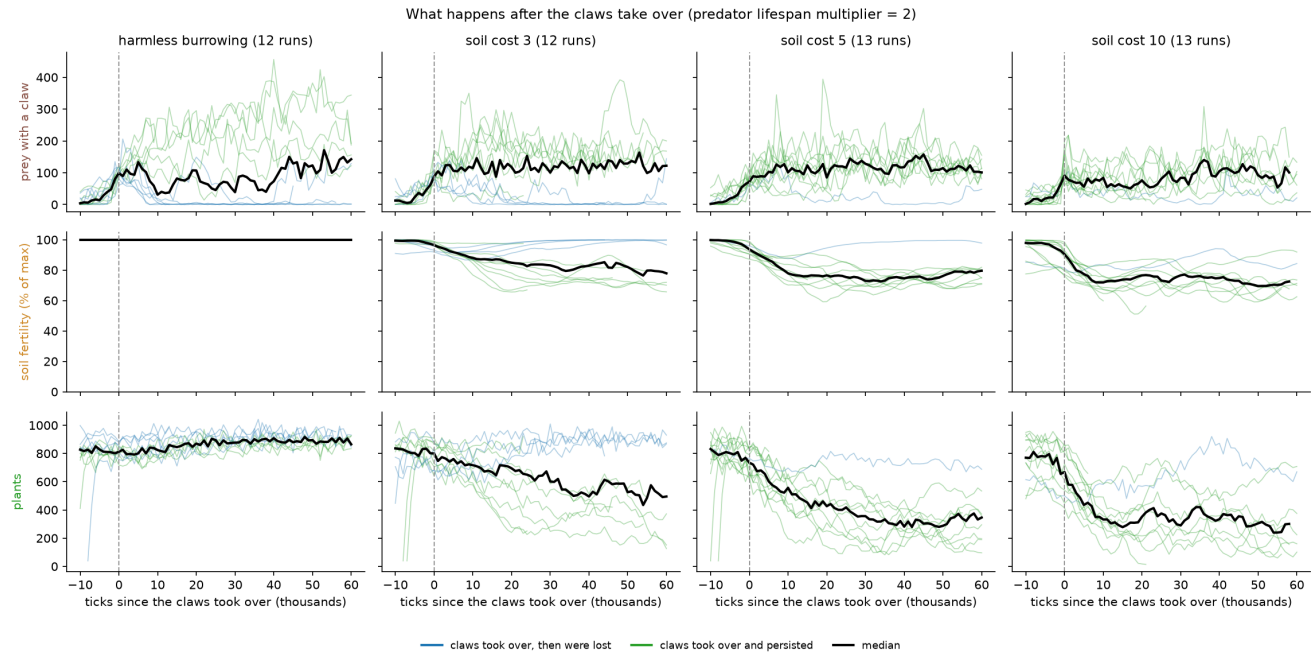


Figure 5. Reference settings with regrowth at 1%, every run lined up on the tick the claws took over (dashed line), median in black. With soil cost 10 the soil drops to a median low of about 73% of its maximum, but the plants level off at a few hundred (median low 350) instead of dying out, so the prey keep eating.

The other changes

With regrowth on, switching off any one of the other three changes still gave no collapses (0 of 20 each), because regrowth alone prevents them. With regrowth off, each change could be measured against the branch with regrowth off and the other three changes on (4 of 20 collapses):

- **Dead plants leaving no food** matters. Giving dead plants back their food raised collapses to 12 of 20 ($p = 0.02$). With less food, prey populations stay smaller, the claw spreads less and the soil takes less damage. With regrowth on, restoring the food also made claws take over more often (17 of 20 runs instead of 13).
- **Predator armour** made no clear difference: 1 of 20 without it ($p = 0.34$).
- **The no-claws-on-plants rule** made no difference: 4 of 20 with claws allowed on plants.
- With all four changes switched off, 9 of 20 runs collapsed.

Reference settings, soil cost 10	Plant regrowth	All prey died	Claws took over
Branch as it is	1%	0 / 20	13 / 20
Predator armour removed	1%	0 / 20	13 / 20
Dead plants leave food	1%	0 / 20	17 / 20
Claws allowed on plants	1%	0 / 20	11 / 20
Other three changes on	off	4 / 20	14 / 20
Predator armour removed	off	1 / 20	11 / 20
Dead plants leave food	off	12 / 20	14 / 20
Claws allowed on plants	off	4 / 20	12 / 20
All four changes off	off	9 / 20	18 / 20

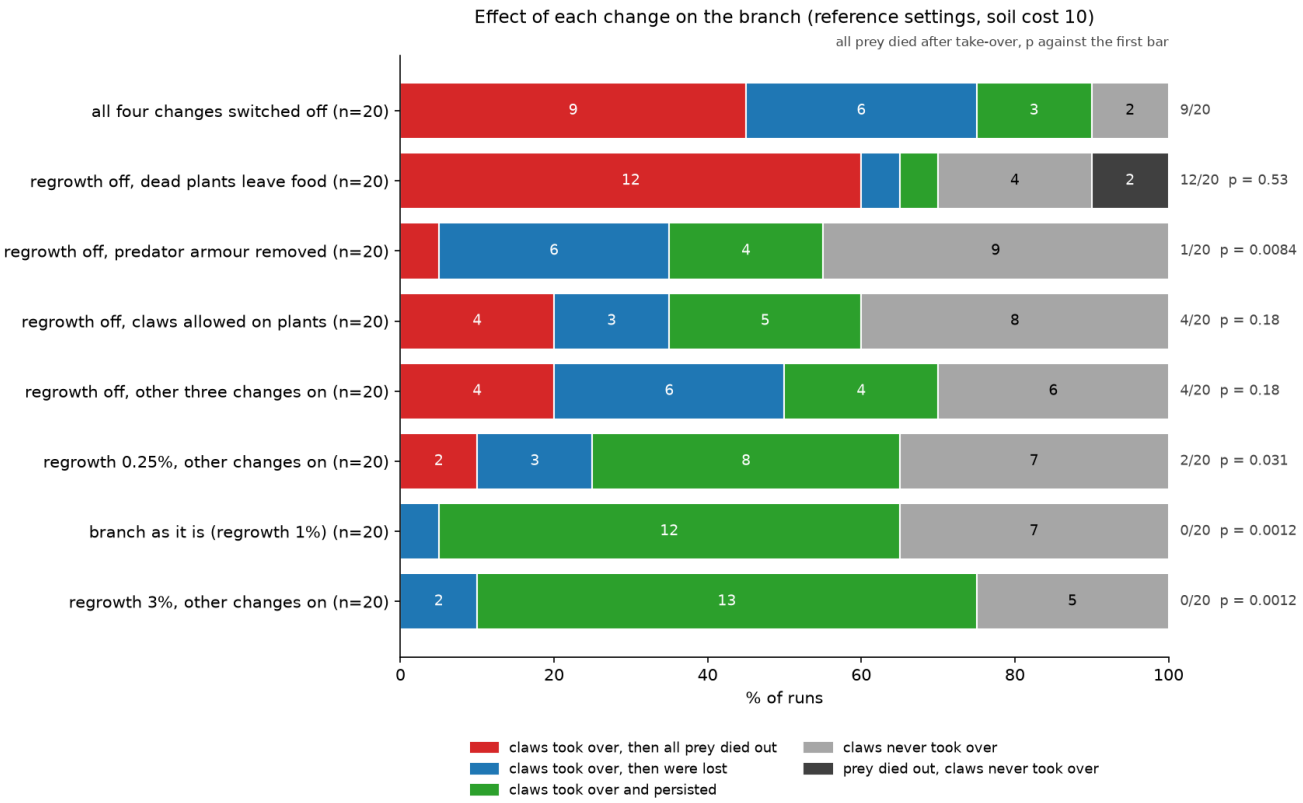


Figure 6. Each change's effect with the reference settings and soil cost 10. The p-values on the right compare each bar with the first (all four changes off); the comparisons in the text use the *regrowth off, other three changes on* bar instead.

Other parameter checks

These used the reference settings with regrowth at 1%. None of them produced a collapse.

Predator lifespan

At predator lifespan multipliers 3, 2, 1.5 and 1 no setup collapsed. At 3, prey without working claws survived only 4 of 8 runs; at the others they survived every run.



Figure 7. Collapses by predator lifespan multiplier and setup (20 runs per cell at 2, 8 at the others).

Hide duration

With 30-tick hides instead of 100, the claw barely spreads: it took over in at most 2 of 20 runs in any setup, and nothing collapsed. Short hides give too little protection and too little time without ageing to pay for the claw cell.

Fine-tuning

Reference settings, regrowth 1%	All prey died	Claws took over	Median lowest soil
As listed above	0 / 20	12 / 20	79%
Hide duration 150	0 / 20	15 / 20	66%
Soil cost 7	0 / 20	12 / 20	76%
Soil regrowth 0.025%	0 / 20	11 / 20	70%
Stay hidden range 3	0 / 20	6 / 20	94%
Stay hidden range 10	0 / 20	17 / 20	61%
Harmless, hide 150 / range 3 / range 10	0 / 20 each	14, 8 and 18 / 20	100%

The stay hidden range is the strongest lever on the claw: the longer prey stay underground while a predator is near, the more the claw pays off and the further it spreads. At range 10 the soil falls furthest, but with regrowth on the plants still hold.

Suggestions for the presentation

- **To show evolutionary suicide, switch plant regrowth off or keep it at 0.25% or below.** The run settings with regrowth off give 10 collapses in 20 runs against none with harmless burrowing ($p = 0.0004$).
- **Plant regrowth is a result in its own right.** A steady outside supply of plants rescues a population from a trait that would otherwise destroy it, which matches the idea that immigration or outside resources can prevent evolutionary suicide. Figure 6 shows this step by step.
- **Choose the plant-death behaviour deliberately.** Whether dead plants leave food changes how often the population collapses.
- **Present the run's graph as a near-collapse and recovery,** and explain the chart's soil scale: the whole-grid average can't go above about 54%, and degraded tiles start near 46%.
- **The stay hidden range** is the simplest setting for making the claw spread more or less.

Limitations

- The rates come from headless runs of the branch's code; they haven't been checked against browser runs.
- 20 runs per setup (8 in the predator check). Differences of 2 to 4 collapses could be chance, so several results are reported as directions rather than proven effects.
- Each grid size gives one soil layout (soil seed 42); only the placement of the starting organisms varied.
- The original run's graph is a single run. It was compared with 20 runs of its settings, not reproduced exactly; the simulation is random beyond the starting layout.
- Armour, the plant-claw rule and the plant-death change were switched off in memory by the test runner. With all four off, the results match the code before the branch's commit.

Reproducing the results

On the jenna-testing branch, with the scripts in `analysis/headless/`:

```
for s in evo_suicide jenna_config regrowth_sweep ablation_sweep ablation_no_regrowth \
         hide_sweep predator_sweep tuning_sweep; do
  node analysis/headless/sweep.js analysis/headless/sweeps/$s.json \
    --out analysis/headless/results/jenna-testing/$s
done

python analysis/headless/plot_evo_suicide.py analysis/headless/results/jenna-testing/jenna_config \
  --rename "her settings=run settings" --damage "run settings" \
  --control "run settings, harmless burrowing" --baseline "run settings, no burrowing" \
  --browser-view "run settings" --browser-view "run settings, no plant regrowth"
```

The runs take about 5 to 6 hours on an 8-core laptop. `plot_steps.py` draws Figure 6 from any chosen setups across result folders.